Model 0

This section presents the R outputs corresponding to Model 0. The following presents the summary of Model 0.

```
Family: gaussian
Links: mu = identity
Formula: log_consumption_std ~ 1 + (1 | nace_section_letter) + (1 | Locality_id)
Data: db (Number of observations: 55374)
Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
total post-warmup draws = 8000

Multilevel Hyperparameters:
 ~Locality_id (Number of levels: 68)
                Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)      0.16       0.02      0.14      0.20  1.00      1379      2147

~nace_section_letter (Number of levels: 20)
                Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)      0.32       0.05      0.24      0.45  1.00      1590      2873

Regression Coefficients:
            Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
Intercept       0.05       0.08     -0.10      0.21  1.01       735      1420

Further Distributional Parameters:
        Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sigma       0.94       0.00      0.94      0.95  1.00     15922      5525

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

The following summarises key descriptive statistics of the posterior distribution for the parameters of Model 0.

Summary of Posterior Distribution

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
(Intercept)	0.05	0.05	0.04	[-0.06, 0.19]	74.65%	[-0.10, 0.10]	74.45%	1.005	702.00

The following results present the estimated intra-class correlation coefficients (ICCs), indicating the proportion of variance attributable to sector, locality and their combined effects.

posterior_summary(icc_nace)

mean	median	l95.2.5%	u95.97.5%
0.10471700	0.09968390	0.05750054	0.18242986

posterior_summary(icc_loc)

mean	median	l95.2.5%	u95.97.5%
0.02672394	0.02613921	0.01797002	0.03872883

posterior_summary(icc_total)

mean	median	l95.2.5%	u95.97.5%
0.13144094	0.12686827	0.08383604	0.20732756

Model 1A

This section presents the R outputs corresponding to Model 1A. The following presents the summary of Model 1A.

Family: gaussian
Links: mu = identity
Formula: log_consumption_std ~ Year + Company_Size_z + (1 | nace_section_letter) + (1 | Locality_id)
Data: db (Number of observations: 55374)
Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
total post-warmup draws = 8000

Multilevel Hyperparameters:
~Locality_id (Number of levels: 68)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.16	0.02	0.14	0.20	1.00	1167	1722

~nace_section_letter (Number of levels: 20)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.32	0.06	0.23	0.44	1.00	1499	2942

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	16.97	5.84	5.31	28.72	1.00	15351	5552
Year	-0.01	0.00	-0.01	-0.00	1.00	15429	5635
Company_Size_z	0.13	0.00	0.12	0.14	1.00	14383	5601

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.93	0.00	0.93	0.94	1.00	14878	5341

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

The following summarise key descriptive statistics of the posterior distribution for the parameters of Model 1A.

Summary of Posterior Distribution

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
(Intercept)	16.98	16.97	17.83	[7.48, 26.09]	99.78%	[-0.10, 0.10]	0%	1.000	15228.00
Year	-8.39e-03	-8.38e-03	-8.79e-03	[-0.01, 0.00]	99.76%	[-0.10, 0.10]	100%	1.000	15298.00
Company_Size_z	0.13	0.13	0.13	[0.13, 0.14]	100%	[-0.10, 0.10]	0%	1.000	14685.00

Fixed effects sigma

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
sigma	0.93	0.93	0.93	[0.93, 0.94]	100%	[-0.10, 0.10]	0%	1.000	14713.00

Random effects Intercept: Locality_id

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
Locality_id.L01	0.03	0.03	0.03	[-0.01, 0.07]	85.46%	[-0.10, 0.10]	100%	1.001	1327.00
Locality_id.L02	0.31	0.31	0.31	[0.25, 0.36]	100%	[-0.10, 0.10]	0%	1.000	2141.00
Locality_id.L03	-0.08	-0.08	-0.08	[-0.16, 0.00]	93.75%	[-0.10, 0.10]	64.53%	1.000	3846.00
Locality_id.L04	-0.08	-0.08	-0.08	[-0.15, -0.02]	97.85%	[-0.10, 0.10]	70.74%	1.001	2775.00
Locality_id.L05	-0.02	-0.02	-0.02	[-0.10, 0.07]	63.91%	[-0.10, 0.10]	97.16%	1.000	4053.00
Locality_id.L06	0.15	0.15	0.16	[0.10, 0.20]	100%	[-0.10, 0.10]	3.58%	1.001	2157.00
Locality_id.L07	-0.02	-0.02	-0.02	[-0.11, 0.06]	66.97%	[-0.10, 0.10]	93.86%	1.000	5236.00
Locality_id.L08	-0.08	-0.08	-0.07	[-0.16, 0.01]	92.30%	[-0.10, 0.10]	66.11%	1.000	4386.00
Locality_id.L09	5.99e-03	6.30e-03	4.14e-03	[-0.08, 0.10]	54.56%	[-0.10, 0.10]	98.09%	1.000	4855.00
Locality_id.L10	-0.05	-0.05	-0.05	[-0.12, 0.03]	84.08%	[-0.10, 0.10]	89.96%	1.000	3860.00
Locality_id.L11	0.25	0.25	0.24	[0.13, 0.35]	99.95%	[-0.10, 0.10]	0%	1.000	6471.00
Locality_id.L12	0.24	0.24	0.24	[0.15, 0.35]	100%	[-0.10, 0.10]	0%	1.000	5715.00

Locality_id.L13	0.17	0.17	0.17	[0.12, 0.23]	100%	[-0.10, 0.10]	0%	1.001	2177.00
Locality_id.L14	-0.12	-0.12	-0.12	[-0.16, -0.08]	100%	[-0.10, 0.10]	22.82%	1.001	1249.00
Locality_id.L15	-0.03	-0.03	-0.02	[-0.13, 0.07]	70.23%	[-0.10, 0.10]	87.79%	1.000	5751.00
Locality_id.L16	-0.29	-0.29	-0.28	[-0.36, -0.22]	100%	[-0.10, 0.10]	0%	1.001	3170.00
Locality_id.L17	-0.16	-0.16	-0.15	[-0.21, -0.10]	100%	[-0.10, 0.10]	3.49%	1.001	2374.00
Locality_id.L18	-0.17	-0.17	-0.16	[-0.21, -0.11]	100%	[-0.10, 0.10]	0%	1.000	1833.00
Locality_id.L19	-0.06	-0.07	-0.07	[-0.15, 0.02]	88.60%	[-0.10, 0.10]	75.30%	1.000	4674.00
Locality_id.L20	-0.21	-0.21	-0.20	[-0.27, -0.15]	100%	[-0.10, 0.10]	0%	1.001	2471.00
Locality_id.L21	0.10	0.10	0.10	[-0.06, 0.26]	84.08%	[-0.10, 0.10]	50.34%	1.000	9531.00
Locality_id.L22	-0.23	-0.23	-0.22	[-0.31, -0.14]	100%	[-0.10, 0.10]	0%	1.000	5337.00
Locality_id.L23	0.08	0.08	0.08	[0.03, 0.14]	99.08%	[-0.10, 0.10]	74.33%	1.001	2005.00
Locality_id.L24	-0.14	-0.14	-0.14	[-0.20, -0.10]	100%	[-0.10, 0.10]	6.76%	1.001	1886.00
Locality_id.L25	0.06	0.06	0.06	[-0.04, 0.17]	82.39%	[-0.10, 0.10]	73.28%	1.000	6400.00
Locality_id.L26	0.23	0.23	0.22	[0.18, 0.28]	100%	[-0.10, 0.10]	0%	1.001	1726.00
Locality_id.L27	-0.02	-0.02	-0.02	[-0.08, 0.03]	75.31%	[-0.10, 0.10]	100%	1.000	2083.00
Locality_id.L28	-0.07	-0.07	-0.06	[-0.11, -0.02]	99.17%	[-0.10, 0.10]	90.21%	1.001	1516.00
Locality_id.L29	0.06	0.06	0.06	[-0.04, 0.15]	84.06%	[-0.10, 0.10]	77.04%	1.000	5526.00
Locality_id.L30	-0.05	-0.05	-0.05	[-0.18, 0.08]	72.34%	[-0.10, 0.10]	75.12%	1.000	9030.00
Locality_id.L31	-0.20	-0.20	-0.21	[-0.29, -0.11]	99.98%	[-0.10, 0.10]	1.29%	1.000	5150.00
Locality_id.L32	0.03	0.03	0.03	[-0.07, 0.12]	67.99%	[-0.10, 0.10]	91.37%	1.000	5273.00
Locality_id.L33	0.06	0.06	0.07	[0.01, 0.12]	97.82%	[-0.10, 0.10]	88.67%	1.001	1844.00
Locality_id.L34	0.02	0.02	0.02	[-0.04, 0.07]	69.10%	[-0.10, 0.10]	100%	1.001	1960.00
Locality_id.L35	-0.26	-0.26	-0.26	[-0.32, -0.21]	100%	[-0.10, 0.10]	0%	1.001	1994.00
Locality_id.L36	-0.17	-0.17	-0.17	[-0.23, -0.11]	100%	[-0.10, 0.10]	0.72%	1.000	2545.00
Locality_id.L37	0.06	0.06	0.06	[-0.01, 0.13]	91.97%	[-0.10, 0.10]	84.08%	1.000	3605.00
Locality_id.L38	-0.06	-0.06	-0.06	[-0.14, 0.01]	91.36%	[-0.10, 0.10]	79.13%	1.000	4100.00
Locality_id.L39	0.27	0.27	0.27	[0.20, 0.34]	100%	[-0.10, 0.10]	0%	1.000	3142.00
Locality_id.L40	0.14	0.14	0.13	[0.02, 0.28]	95.14%	[-0.10, 0.10]	31.86%	1.000	8917.00
Locality_id.L41	-0.10	-0.10	-0.09	[-0.17, -0.02]	97.70%	[-0.10, 0.10]	52.74%	1.001	4030.00
Locality_id.L42	0.18	0.18	0.18	[0.13, 0.23]	100%	[-0.10, 0.10]	0%	1.001	2013.00
Locality_id.L43	-0.05	-0.05	-0.05	[-0.11, 0.02]	88.34%	[-0.10, 0.10]	91.18%	1.000	2992.00
Locality_id.L44	-0.32	-0.32	-0.32	[-0.43, -0.21]	100%	[-0.10, 0.10]	0%	1.001	6743.00
Locality_id.L45	-0.10	-0.10	-0.11	[-0.23, 0.02]	90.21%	[-0.10, 0.10]	48.34%	1.000	9577.00

Locality_id.L46	0.06	0.06	0.07	[-0.03, 0.15]	85.01%	[-0.10, 0.10]	76.99%	1.000	4924.00
Locality_id.L47	0.17	0.16	0.17	[0.06, 0.27]	99.28%	[-0.10, 0.10]	15.16%	1.000	6234.00
Locality_id.L48	-0.02	-0.02	-0.02	[-0.09, 0.05]	69.20%	[-0.10, 0.10]	97.71%	1.001	3470.00
Locality_id.L49	-0.04	-0.05	-0.04	[-0.13, 0.04]	80.08%	[-0.10, 0.10]	85.67%	1.000	4809.00
Locality_id.L50	-0.02	-0.02	-0.02	[-0.07, 0.03]	71.15%	[-0.10, 0.10]	100%	1.000	1966.00
Locality_id.L51	-0.16	-0.16	-0.16	[-0.29, -0.03]	97.50%	[-0.10, 0.10]	20.97%	1.000	8870.00
Locality_id.L52	-0.17	-0.17	-0.17	[-0.27, -0.06]	99.61%	[-0.10, 0.10]	12.08%	1.000	6119.00
Locality_id.L53	-0.09	-0.09	-0.09	[-0.15, -0.03]	98.94%	[-0.10, 0.10]	62.33%	1.001	2733.00
Locality_id.L54	-3.33e-03	-2.48e-03	-7.63e-03	[-0.07, 0.07]	52.59%	[-0.10, 0.10]	100%	1.000	3597.00
Locality_id.L55	0.36	0.36	0.35	[0.25, 0.47]	100%	[-0.10, 0.10]	0%	1.000	7053.00
Locality_id.L56	0.09	0.09	0.09	[0.04, 0.14]	99.62%	[-0.10, 0.10]	63.30%	1.000	1874.00
Locality_id.L57	0.22	0.22	0.22	[0.18, 0.27]	100%	[-0.10, 0.10]	0%	1.001	1772.00
Locality_id.L58	0.04	0.04	0.04	[-0.01, 0.09]	89.92%	[-0.10, 0.10]	99.57%	1.001	1820.00
Locality_id.L59	-0.08	-0.08	-0.08	[-0.23, 0.07]	80.75%	[-0.10, 0.10]	58.28%	1.000	10470.00
Locality_id.L60	-0.06	-0.06	-0.07	[-0.11, -0.02]	99.17%	[-0.10, 0.10]	93.09%	1.001	1336.00
Locality_id.L61	-0.25	-0.25	-0.24	[-0.37, -0.13]	99.92%	[-0.10, 0.10]	0%	1.000	6937.00
Locality_id.L62	-0.13	-0.13	-0.14	[-0.19, -0.08]	100%	[-0.10, 0.10]	16.80%	1.000	2375.00
Locality_id.L63	0.22	0.22	0.21	[0.11, 0.33]	99.89%	[-0.10, 0.10]	2.16%	1.000	6758.00
Locality_id.L64	0.23	0.23	0.22	[0.13, 0.34]	99.99%	[-0.10, 0.10]	0.09%	1.000	5590.00
Locality_id.L65	0.20	0.20	0.20	[0.12, 0.28]	99.99%	[-0.10, 0.10]	0.13%	1.000	3930.00
Locality_id.L66	0.05	0.05	0.05	[-0.02, 0.12]	87.66%	[-0.10, 0.10]	87.39%	1.000	3151.00
Locality_id.L67	0.09	0.09	0.09	[0.05, 0.14]	99.90%	[-0.10, 0.10]	62.66%	1.000	1509.00
Locality_id.L68	0.03	0.03	0.03	[-0.02, 0.07]	82.39%	[-0.10, 0.10]	100%	1.001	1384.00

Random effects Intercept: nace_section_letter

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
nace_section_letter.A	-0.33	-0.33	-0.32	[-0.44, -0.21]	100%	[-0.10, 0.10]	0%	1.002	993.00
nace_section_letter.B	-0.07	-0.07	-0.07	[-0.25, 0.10]	75.50%	[-0.10, 0.10]	56.51%	1.001	2280.00
nace_section_letter.C	0.18	0.18	0.19	[0.06, 0.29]	99.06%	[-0.10, 0.10]	10.97%	1.002	952.00
nace_section_letter.D	0.06	0.06	0.06	[-0.09, 0.19]	74.40%	[-0.10, 0.10]	68.16%	1.002	1349.00
nace_section_letter.E	0.40	0.40	0.40	[0.27, 0.54]	99.99%	[-0.10, 0.10]	0%	1.001	1373.00

nace_section_letter.F	-0.58	-0.58	-0.57	[-0.70, -0.47]	100%	[-0.10, 0.10]	0%	1.002	997.00
nace_section_letter.G	-0.14	-0.14	-0.13	[-0.26, -0.03]	97.52%	[-0.10, 0.10]	26.25%	1.002	947.00
nace_section_letter.H	-0.33	-0.33	-0.33	[-0.44, -0.21]	100%	[-0.10, 0.10]	0%	1.002	1027.00
nace_section_letter.I	0.52	0.52	0.52	[0.41, 0.64]	100%	[-0.10, 0.10]	0%	1.002	984.00
nace_section_letter.J	0.40	0.40	0.40	[0.28, 0.51]	100%	[-0.10, 0.10]	0%	1.002	1018.00
nace_section_letter.K	0.26	0.26	0.27	[0.14, 0.38]	99.98%	[-0.10, 0.10]	0%	1.002	1010.00
nace_section_letter.L	0.08	0.08	0.08	[-0.04, 0.19]	85.24%	[-0.10, 0.10]	64.08%	1.002	992.00
nace_section_letter.M	-0.28	-0.29	-0.29	[-0.40, -0.17]	99.99%	[-0.10, 0.10]	0%	1.002	992.00
nace_section_letter.N	-0.28	-0.28	-0.28	[-0.40, -0.16]	99.99%	[-0.10, 0.10]	0%	1.002	1010.00
nace_section_letter.O	0.09	0.09	0.09	[-0.03, 0.20]	88.11%	[-0.10, 0.10]	57.03%	1.002	1010.00
nace_section_letter.P	0.05	0.05	0.05	[-0.07, 0.17]	73.46%	[-0.10, 0.10]	76.00%	1.002	1142.00
nace_section_letter.Q	-0.04	-0.04	-0.04	[-0.17, 0.08]	70.93%	[-0.10, 0.10]	77.76%	1.001	1088.00
nace_section_letter.R	0.18	0.18	0.18	[0.06, 0.30]	98.84%	[-0.10, 0.10]	12.74%	1.001	1090.00
nace_section_letter.S	-0.37	-0.37	-0.36	[-0.49, -0.26]	100%	[-0.10, 0.10]	0%	1.002	1008.00
nace_section_letter.T	0.19	0.19	0.18	[-0.16, 0.54]	80.96%	[-0.10, 0.10]	26.09%	1.000	8709.00

Random effects SD/Cor: Locality_id

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
(Intercept)	0.16	0.16	0.16	[0.14, 0.19]	100%	[-0.10, 0.10]	0%	1.003	1145.00

Random effects SD/Cor: nace_section_letter

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
(Intercept)	0.31	0.32	0.29	[0.23, 0.40]	100%	[-0.10, 0.10]	0%	1.001	1575.00

Model 1B

This section presents the R outputs corresponding to Model 1B. The following presents the summary of Model 1B.

Family: gaussian
Links: mu = identity
Formula: log_consumption_std ~ Year + Company_Size_z + (1 | nace_section_letter) + (1 | Locality_id)
Data: db (Number of observations: 55374)
Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
total post-warmup draws = 8000

Multilevel Hyperparameters:
~Locality_id (Number of levels: 68)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.16	0.02	0.14	0.20	1.00	1348	2151

~nace_section_letter (Number of levels: 20)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.32	0.06	0.23	0.45	1.00	1402	3053

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	17.04	5.91	5.44	28.41	1.00	14709	5373
Year	-0.01	0.00	-0.01	-0.00	1.00	14793	5356
Company_Size_z	0.13	0.00	0.12	0.14	1.00	12951	5908

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.93	0.00	0.93	0.94	1.00	14949	5776

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

The following summarise key descriptive statistics of the posterior distribution for the parameters of Model 1B.

Summary of Posterior Distribution

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
(Intercept)	17.03	17.04	16.02	[8.10, 26.87]	99.84%	[-0.10, 0.10]	0%	1.000	14434.00
Year	-8.41e-03	-8.41e-03	-7.89e-03	[-0.01, 0.00]	99.84%	[-0.10, 0.10]	100%	1.000	14523.00
Company_Size_z	0.13	0.13	0.13	[0.13, 0.14]	100%	[-0.10, 0.10]	0%	1.000	13365.00

Fixed effects sigma

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
sigma	0.93	0.93	0.93	[0.93, 0.94]	100%	[-0.10, 0.10]	0%	1.000	14958.00

Random effects Intercept: Locality_id

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
Locality_id.L01	0.03	0.03	0.03	[-0.01, 0.07]	83.08%	[-0.10, 0.10]	100%	1.012	1093.00
Locality_id.L02	0.31	0.31	0.31	[0.25, 0.37]	100%	[-0.10, 0.10]	0%	1.005	1861.00
Locality_id.L03	-0.08	-0.08	-0.09	[-0.17, 0.00]	94.15%	[-0.10, 0.10]	63.13%	1.003	3576.00
Locality_id.L04	-0.08	-0.08	-0.08	[-0.14, -0.02]	98.01%	[-0.10, 0.10]	68.50%	1.005	2211.00
Locality_id.L05	-0.02	-0.02	-0.02	[-0.10, 0.07]	62.89%	[-0.10, 0.10]	96.16%	1.002	3511.00
Locality_id.L06	0.15	0.15	0.15	[0.10, 0.21]	100%	[-0.10, 0.10]	4.55%	1.005	1730.00
Locality_id.L07	-0.03	-0.03	-0.02	[-0.11, 0.06]	67.81%	[-0.10, 0.10]	93.93%	1.002	4200.00
Locality_id.L08	-0.08	-0.08	-0.07	[-0.17, 0.01]	92.10%	[-0.10, 0.10]	65.53%	1.002	3137.00
Locality_id.L09	4.62e-03	5.70e-03	-1.10e-03	[-0.08, 0.10]	53.45%	[-0.10, 0.10]	97.37%	1.001	4233.00
Locality_id.L10	-0.05	-0.05	-0.05	[-0.12, 0.03]	84.88%	[-0.10, 0.10]	89.41%	1.003	2670.00
Locality_id.L11	0.25	0.25	0.25	[0.13, 0.36]	99.94%	[-0.10, 0.10]	0%	1.001	5969.00
Locality_id.L12	0.24	0.24	0.25	[0.14, 0.34]	100%	[-0.10, 0.10]	0%	1.001	6073.00
Locality_id.L13	0.17	0.17	0.17	[0.12, 0.23]	100%	[-0.10, 0.10]	0%	1.005	1861.00

Locality_id.L14	-0.12	-0.12	-0.12	[-0.16, -0.08]	100%	[-0.10, 0.10]	20.71%	1.011	1083.00
Locality_id.L15	-0.03	-0.03	-0.04	[-0.13, 0.06]	71.10%	[-0.10, 0.10]	88.00%	1.002	4578.00
Locality_id.L16	-0.29	-0.29	-0.29	[-0.37, -0.22]	100%	[-0.10, 0.10]	0%	1.004	2924.00
Locality_id.L17	-0.16	-0.16	-0.17	[-0.21, -0.10]	100%	[-0.10, 0.10]	4.17%	1.006	1845.00
Locality_id.L18	-0.17	-0.17	-0.16	[-0.21, -0.11]	100%	[-0.10, 0.10]	0%	1.007	1504.00
Locality_id.L19	-0.07	-0.07	-0.06	[-0.15, 0.02]	89.34%	[-0.10, 0.10]	74.11%	1.002	3894.00
Locality_id.L20	-0.21	-0.21	-0.21	[-0.27, -0.15]	100%	[-0.10, 0.10]	0%	1.004	1933.00
Locality_id.L21	0.10	0.10	0.09	[-0.07, 0.25]	83.50%	[-0.10, 0.10]	51.72%	1.000	9144.00
Locality_id.L22	-0.23	-0.23	-0.22	[-0.31, -0.14]	100%	[-0.10, 0.10]	0%	1.002	3782.00
Locality_id.L23	0.08	0.08	0.08	[0.03, 0.13]	98.91%	[-0.10, 0.10]	75.72%	1.006	1639.00
Locality_id.L24	-0.14	-0.14	-0.14	[-0.20, -0.10]	100%	[-0.10, 0.10]	6.29%	1.008	1534.00
Locality_id.L25	0.06	0.06	0.06	[-0.05, 0.16]	80.44%	[-0.10, 0.10]	74.54%	1.001	6573.00
Locality_id.L26	0.23	0.23	0.23	[0.18, 0.28]	100%	[-0.10, 0.10]	0%	1.008	1339.00
Locality_id.L27	-0.02	-0.03	-0.02	[-0.08, 0.03]	77.30%	[-0.10, 0.10]	100%	1.006	1635.00
Locality_id.L28	-0.07	-0.07	-0.07	[-0.11, -0.02]	99.21%	[-0.10, 0.10]	88.63%	1.009	1217.00
Locality_id.L29	0.06	0.06	0.07	[-0.04, 0.15]	83.67%	[-0.10, 0.10]	77.70%	1.002	4340.00
Locality_id.L30	-0.05	-0.05	-0.05	[-0.17, 0.08]	72.80%	[-0.10, 0.10]	76.37%	1.001	6808.00
Locality_id.L31	-0.21	-0.20	-0.21	[-0.30, -0.12]	99.92%	[-0.10, 0.10]	0.76%	1.002	4059.00
Locality_id.L32	0.03	0.03	0.03	[-0.07, 0.12]	67.19%	[-0.10, 0.10]	91.34%	1.002	4705.00
Locality_id.L33	0.06	0.06	0.06	[0.01, 0.11]	97.09%	[-0.10, 0.10]	89.76%	1.007	1577.00
Locality_id.L34	0.02	0.02	8.09e-03	[-0.04, 0.07]	68.15%	[-0.10, 0.10]	100%	1.006	1627.00
Locality_id.L35	-0.26	-0.26	-0.26	[-0.31, -0.21]	100%	[-0.10, 0.10]	0%	1.007	1618.00
Locality_id.L36	-0.17	-0.17	-0.17	[-0.23, -0.11]	99.99%	[-0.10, 0.10]	1.05%	1.005	2225.00
Locality_id.L37	0.06	0.06	0.06	[-0.01, 0.13]	91.46%	[-0.10, 0.10]	84.89%	1.004	2510.00
Locality_id.L38	-0.07	-0.07	-0.07	[-0.14, 0.01]	90.95%	[-0.10, 0.10]	77.07%	1.003	3486.00
Locality_id.L39	0.27	0.27	0.27	[0.20, 0.34]	100%	[-0.10, 0.10]	0%	1.003	2516.00
Locality_id.L40	0.13	0.13	0.14	[0.00, 0.26]	94.80%	[-0.10, 0.10]	33.62%	1.001	7267.00
Locality_id.L41	-0.10	-0.10	-0.10	[-0.17, -0.02]	98.01%	[-0.10, 0.10]	50.88%	1.002	3370.00
Locality_id.L42	0.18	0.18	0.17	[0.12, 0.23]	100%	[-0.10, 0.10]	0%	1.008	1533.00
Locality_id.L43	-0.05	-0.05	-0.05	[-0.11, 0.02]	89.22%	[-0.10, 0.10]	90.62%	1.004	2450.00
Locality_id.L44	-0.32	-0.32	-0.32	[-0.43, -0.21]	100%	[-0.10, 0.10]	0%	1.001	6802.00
Locality_id.L45	-0.10	-0.10	-0.11	[-0.23, 0.02]	90.59%	[-0.10, 0.10]	47.89%	1.001	7163.00
Locality_id.L46	0.06	0.06	0.06	[-0.03, 0.15]	85.20%	[-0.10, 0.10]	77.51%	1.001	4515.00

Locality_id.L47	0.16	0.16	0.17	[0.06, 0.26]	99.35%	[-0.10, 0.10]	14.12%	1.002	4906.00
Locality_id.L48	-0.02	-0.02	-0.02	[-0.09, 0.05]	71.25%	[-0.10, 0.10]	97.88%	1.003	2817.00
Locality_id.L49	-0.05	-0.05	-0.05	[-0.13, 0.04]	80.34%	[-0.10, 0.10]	84.70%	1.001	4151.00
Locality_id.L50	-0.02	-0.02	-0.01	[-0.07, 0.04]	71.64%	[-0.10, 0.10]	100%	1.007	1610.00
Locality_id.L51	-0.16	-0.16	-0.15	[-0.30, -0.03]	97.44%	[-0.10, 0.10]	21.05%	1.001	7604.00
Locality_id.L52	-0.17	-0.17	-0.17	[-0.28, -0.07]	99.61%	[-0.10, 0.10]	11.66%	1.002	4650.00
Locality_id.L53	-0.09	-0.09	-0.09	[-0.15, -0.03]	98.95%	[-0.10, 0.10]	60.76%	1.005	2092.00
Locality_id.L54	-4.20e-03	-4.04e-03	-3.05e-03	[-0.08, 0.07]	53.65%	[-0.10, 0.10]	100%	1.003	2800.00
Locality_id.L55	0.36	0.36	0.35	[0.25, 0.47]	100%	[-0.10, 0.10]	0%	1.001	5849.00
Locality_id.L56	0.09	0.09	0.08	[0.04, 0.14]	99.58%	[-0.10, 0.10]	65.26%	1.006	1555.00
Locality_id.L57	0.22	0.22	0.21	[0.17, 0.27]	100%	[-0.10, 0.10]	0%	1.007	1306.00
Locality_id.L58	0.04	0.04	0.05	[-0.01, 0.09]	88.76%	[-0.10, 0.10]	100%	1.007	1502.00
Locality_id.L59	-0.08	-0.08	-0.08	[-0.24, 0.06]	80.99%	[-0.10, 0.10]	57.79%	1.000	8308.00
Locality_id.L60	-0.07	-0.07	-0.06	[-0.11, -0.02]	99.16%	[-0.10, 0.10]	91.63%	1.010	1104.00
Locality_id.L61	-0.25	-0.25	-0.25	[-0.38, -0.14]	99.96%	[-0.10, 0.10]	0%	1.001	6134.00
Locality_id.L62	-0.13	-0.13	-0.14	[-0.19, -0.07]	100%	[-0.10, 0.10]	16.57%	1.004	2061.00
Locality_id.L63	0.21	0.21	0.21	[0.11, 0.33]	99.88%	[-0.10, 0.10]	2.34%	1.001	6445.00
Locality_id.L64	0.23	0.23	0.23	[0.13, 0.34]	99.96%	[-0.10, 0.10]	0.29%	1.001	5028.00
Locality_id.L65	0.20	0.20	0.19	[0.12, 0.28]	100%	[-0.10, 0.10]	0.37%	1.003	3599.00
Locality_id.L66	0.05	0.05	0.05	[-0.02, 0.12]	87.08%	[-0.10, 0.10]	88.20%	1.004	2657.00
Locality_id.L67	0.09	0.09	0.09	[0.04, 0.14]	99.86%	[-0.10, 0.10]	64.36%	1.007	1309.00
Locality_id.L68	0.02	0.02	0.02	[-0.02, 0.07]	80.79%	[-0.10, 0.10]	100%	1.010	1060.00

Random effects Intercept: nace_section_letter

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
nace_section_letter.A	-0.33	-0.33	-0.33	[-0.44, -0.21]	100%	[-0.10, 0.10]	0%	1.006	984.00
nace_section_letter.B	-0.07	-0.07	-0.08	[-0.25, 0.10]	74.16%	[-0.10, 0.10]	57.55%	1.003	2102.00
nace_section_letter.C	0.18	0.18	0.19	[0.06, 0.29]	98.94%	[-0.10, 0.10]	10.50%	1.006	973.00
nace_section_letter.D	0.06	0.06	0.06	[-0.08, 0.19]	75.11%	[-0.10, 0.10]	67.57%	1.006	1251.00
nace_section_letter.E	0.41	0.41	0.40	[0.28, 0.55]	99.99%	[-0.10, 0.10]	0%	1.005	1275.00
nace_section_letter.F	-0.58	-0.58	-0.57	[-0.69, -0.46]	100%	[-0.10, 0.10]	0%	1.006	982.00

nace_section_letter.G	-0.14	-0.14	-0.13	[-0.25, -0.02]	97.52%	[-0.10, 0.10]	28.12%	1.006	967.00
nace_section_letter.H	-0.32	-0.32	-0.32	[-0.44, -0.21]	100%	[-0.10, 0.10]	0%	1.006	1025.00
nace_section_letter.I	0.52	0.52	0.52	[0.41, 0.64]	100%	[-0.10, 0.10]	0%	1.006	981.00
nace_section_letter.J	0.40	0.40	0.41	[0.28, 0.51]	99.99%	[-0.10, 0.10]	0%	1.006	978.00
nace_section_letter.K	0.27	0.26	0.26	[0.14, 0.39]	99.89%	[-0.10, 0.10]	0%	1.006	1054.00
nace_section_letter.L	0.08	0.08	0.08	[-0.04, 0.19]	85.88%	[-0.10, 0.10]	62.43%	1.006	999.00
nace_section_letter.M	-0.28	-0.28	-0.28	[-0.41, -0.17]	100%	[-0.10, 0.10]	0%	1.006	1043.00
nace_section_letter.N	-0.28	-0.28	-0.28	[-0.40, -0.16]	100%	[-0.10, 0.10]	0%	1.006	1017.00
nace_section_letter.O	0.09	0.09	0.09	[-0.03, 0.21]	88.36%	[-0.10, 0.10]	55.89%	1.006	1025.00
nace_section_letter.P	0.05	0.05	0.04	[-0.07, 0.18]	74.39%	[-0.10, 0.10]	74.37%	1.006	1084.00
nace_section_letter.Q	-0.04	-0.04	-0.03	[-0.17, 0.08]	69.62%	[-0.10, 0.10]	78.89%	1.005	1094.00
nace_section_letter.R	0.18	0.18	0.19	[0.06, 0.30]	98.54%	[-0.10, 0.10]	12.13%	1.006	1052.00
nace_section_letter.S	-0.36	-0.37	-0.36	[-0.49, -0.25]	100%	[-0.10, 0.10]	0%	1.006	1013.00
nace_section_letter.T	0.20	0.20	0.19	[-0.16, 0.54]	81.61%	[-0.10, 0.10]	25.75%	1.000	6853.00

Random effects SD/Cor: Locality_id

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
(Intercept)	0.16	0.16	0.16	[0.14, 0.19]	100%	[-0.10, 0.10]	0%	1.004	1320.00

Random effects SD/Cor: nace_section_letter

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
(Intercept)	0.31	0.32	0.29	[0.23, 0.40]	100%	[-0.10, 0.10]	0%	1.001	1386.00

Model 1A*

This section presents the R outputs corresponding to Model 1A*. The following presents the summary of Model 1A*.

```
 Family: gaussian
  Links: mu = identity
Formula: log_consumption_std ~ Year + Company_Size_z + (1 | nace_section_letter)
   Data: db (Number of observations: 55374)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 8000

Multilevel Hyperparameters:
~nace_section_letter (Number of levels: 20)
                 Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)      0.33       0.06       0.24      0.47  1.00      1314      2521

Regression Coefficients:
                    Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
Intercept             17.45       5.96      5.70     29.15  1.00      7326      5213
Year                  -0.01       0.00     -0.01     -0.00  1.00      7335      5249
Company_Size_z         0.13       0.00      0.13      0.14  1.00      6864      4774

Further Distributional Parameters:
        Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sigma       0.94       0.00      0.94      0.95  1.00      7442      4931

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

The following table summarises key descriptive statistics of the posterior distribution for the parameters of Model 1A*.

Summary of Posterior Distribution

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
(Intercept)	17.43	17.45	17.37	[7.95, 26.78]	99.81%	[-0.10, 0.10]	0%	1.000	7298.00
Year	-8.60e-03	-8.61e-03	-8.57e-03	[-0.01, 0.00]	99.81%	[-0.10, 0.10]	100%	1.000	7309.00
Company_Size_z	0.13	0.13	0.13	[0.13, 0.14]	100%	[-0.10, 0.10]	0%	1.000	6796.00

Fixed effects sigma

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
sigma	0.94	0.94	0.94	[0.94, 0.95]	100%	[-0.10, 0.10]	0%	1.000	7391.00

Random effects Intercept: nace_section_letter

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
nace_section_letter.A	-0.38	-0.38	-0.38	[-0.50, -0.26]	100%	[-0.10, 0.10]	0%	1.002	1270.00
nace_section_letter.B	-0.09	-0.09	-0.10	[-0.27, 0.09]	77.69%	[-0.10, 0.10]	52.43%	1.001	2467.00
nace_section_letter.C	0.20	0.20	0.20	[0.09, 0.33]	99.62%	[-0.10, 0.10]	6.86%	1.002	1265.00
nace_section_letter.D	0.10	0.10	0.10	[-0.04, 0.24]	87.50%	[-0.10, 0.10]	49.82%	1.002	1714.00
nace_section_letter.E	0.41	0.42	0.42	[0.28, 0.55]	100%	[-0.10, 0.10]	0%	1.001	1710.00
nace_section_letter.F	-0.60	-0.60	-0.60	[-0.72, -0.48]	100%	[-0.10, 0.10]	0%	1.002	1264.00
nace_section_letter.G	-0.15	-0.15	-0.16	[-0.28, -0.04]	98.04%	[-0.10, 0.10]	20.92%	1.002	1241.00
nace_section_letter.H	-0.31	-0.31	-0.31	[-0.44, -0.19]	99.99%	[-0.10, 0.10]	0%	1.002	1292.00
nace_section_letter.I	0.53	0.53	0.52	[0.40, 0.64]	100%	[-0.10, 0.10]	0%	1.002	1254.00
nace_section_letter.J	0.40	0.40	0.39	[0.27, 0.52]	100%	[-0.10, 0.10]	0%	1.002	1272.00
nace_section_letter.K	0.27	0.26	0.27	[0.14, 0.39]	99.96%	[-0.10, 0.10]	0%	1.002	1340.00

nace_section_letter.L	0.11	0.11	0.11	[-0.01, 0.23]	91.75%	[-0.10, 0.10]	46.75%	1.002	1264.00
nace_section_letter.M	-0.28	-0.28	-0.27	[-0.40, -0.15]	99.95%	[-0.10, 0.10]	0%	1.002	1287.00
nace_section_letter.N	-0.29	-0.29	-0.28	[-0.41, -0.16]	99.95%	[-0.10, 0.10]	0%	1.002	1318.00
nace_section_letter.O	0.08	0.08	0.08	[-0.04, 0.21]	84.81%	[-0.10, 0.10]	62.68%	1.002	1292.00
nace_section_letter.P	0.09	0.09	0.08	[-0.05, 0.21]	85.58%	[-0.10, 0.10]	57.68%	1.002	1405.00
nace_section_letter.Q	-0.05	-0.05	-0.06	[-0.18, 0.07]	74.11%	[-0.10, 0.10]	74.49%	1.002	1398.00
nace_section_letter.R	0.19	0.19	0.19	[0.06, 0.31]	99.08%	[-0.10, 0.10]	12.01%	1.002	1338.00
nace_section_letter.S	-0.39	-0.40	-0.39	[-0.52, -0.28]	100%	[-0.10, 0.10]	0%	1.002	1275.00
nace_section_letter.T	0.12	0.12	0.13	[-0.24, 0.47]	71.25%	[-0.10, 0.10]	31.32%	1.000	5578.00

Random effects SD/Cor: nace_section_letter

Parameter	Median	Mean	MAP	89% CI	pd	ROPE	% in ROPE	Rhat	ESS
(Intercept)	0.32	0.33	0.30	[0.24, 0.41]	100%	[-0.10, 0.10]	0%	1.002	1336.00

Model 2A

This section presents the R outputs corresponding to Model 2A. The following presents the summary of Model 2A.

```
Family: gaussian
  Links: mu = identity
Formula: log_consumption_std ~ Year + Company_Size_z + (1 | Company_id) + (1 | nace_section_letter) + (1 | Locality_id)
   Data: db (Number of observations: 55374)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 8000

Multilevel Hyperparameters:
~Company_id (Number of levels: 11706)
                Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)       0.88       0.01      0.87      0.89  1.01       741      1435

~Locality_id (Number of levels: 68)
                Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)       0.13       0.02      0.10      0.17  1.01       522      1103

~nace_section_letter (Number of levels: 20)
                Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)       0.31       0.06      0.22      0.44  1.00      1386      3023

Regression Coefficients:
                     Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
Intercept               18.82       2.10     14.70     22.89  1.00      8872      5604
Year                    -0.01       0.00     -0.01     -0.01  1.00      8962      5565
Company_Size_z           0.18       0.01      0.17      0.20  1.00      1616      2776

Further Distributional Parameters:
      Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sigma     0.33       0.00      0.33      0.34  1.00      8579      6676
```

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

The following summarise key descriptive statistics of the posterior distribution for the parameters of Model 2A. To maintain clarity and interpretability, for the company-level random intercepts, we present the five smallest companies and the five largest companies, based on the company size variable.

Fixed Effects

Parameter	Median	Mean	MAP	89% CI	pd	% in ROPE	Rhat	ESS
(Intercept)	18.81	18.82	18.59	[15.58, 22.20]	100%	0%	1.000	8893.00
Year	-9.29e-03	-9.30e-03	-9.15e-03	[-0.01, -0.01]	100%	100%	1.000	8986.00
Company_Size_z	0.18	0.18	0.18	[0.17, 0.19]	100%	0%	1.002	1593.00

Sigma

Parameter	Median	Mean	MAP	89% CI	pd	% in ROPE	Rhat	ESS
sigma	0.33	0.33	0.33	[0.33, 0.34]	100%	0%	1.000	8416.00

Random Effects (Count Model): Company_id

Parameter	Median	Mean	MAP	89% CI	pd	% in ROPE	Rhat	ESS
Company_id.5	0.52	0.52	0.52	[0.23, 0.78]	99.89%	0%	1.000	2500.00
Company_id.7	-0.94	-0.94	-0.95	[-1.22, -0.69]	100%	0%	1.003	1998.00
Company_id.48	-0.23	-0.23	-0.21	[-0.50, 0.03]	91.51%	21.00%	1.002	2005.00

Company_id.49	1.55	1.55	1.53	[1.29, 1.84]	100%	0%	1.000	3470.00
Company_id.62	-0.44	-0.44	-0.45	[-0.70, -0.18]	99.55%	0%	1.001	1669.00
Company_id.49838	-0.70	-0.70	-0.70	[-1.01, -0.41]	99.99%	0%	1.003	1629.00
Company_id.24514	1.02	1.01	1.03	[0.67, 1.35]	100%	0%	1.002	1571.00
Company_id.44233	-3.04	-3.04	-3.02	[-3.39, -2.69]	100%	0%	1.002	2097.00
Company_id.12308	-1.39	-1.38	-1.39	[-1.77, -1.00]	100%	0%	1.002	1626.00
Company_id.58635	-14.48	-14.47	-14.49	[-15.51, -13.39]	100%	0%	1.002	1710.00
(Intercept)	0.88	0.88	0.88	[0.87, 0.89]	100%	0%	1.006	734.00

Random Effects (Count Model): Locality_id

Parameter	Median	Mean	MAP	89% CI	pd	% in ROPE	Rhat	ESS
Locality_id.L01	0.02	0.02	0.02	[-0.05, 0.08]	67.29%	100%	1.011	343.00
Locality_id.L02	0.27	0.27	0.26	[0.17, 0.36]	100%	0%	1.015	329.00
Locality_id.L03	-0.05	-0.05	-0.05	[-0.18, 0.08]	72.36%	74.66%	1.005	774.00
Locality_id.L04	-0.07	-0.07	-0.06	[-0.17, 0.04]	84.67%	70.42%	1.004	515.00
Locality_id.L05	-6.51e-03	-7.84e-03	-6.51e-03	[-0.14, 0.12]	53.31%	82.33%	1.006	598.00
Locality_id.L06	0.12	0.12	0.13	[0.03, 0.21]	98.49%	31.79%	1.007	437.00
Locality_id.L07	-0.02	-0.02	-0.01	[-0.18, 0.11]	58.63%	77.04%	1.005	631.00
Locality_id.L08	-0.04	-0.04	-0.04	[-0.16, 0.11]	67.11%	75.45%	1.006	671.00
Locality_id.L09	-0.02	-0.02	-8.06e-03	[-0.16, 0.12]	57.06%	77.89%	1.002	763.00
Locality_id.L10	-0.03	-0.03	-0.03	[-0.15, 0.08]	68.47%	82.13%	1.008	601.00
Locality_id.L11	0.10	0.10	0.10	[-0.07, 0.26]	84.36%	49.04%	1.005	924.00
Locality_id.L12	0.15	0.15	0.15	[-0.01, 0.30]	93.12%	30.28%	1.004	707.00
Locality_id.L13	0.15	0.15	0.15	[0.07, 0.24]	99.83%	13.76%	1.003	402.00
Locality_id.L14	-0.12	-0.12	-0.12	[-0.18, -0.06]	99.92%	26.28%	1.019	407.00
Locality_id.L15	9.96e-03	8.89e-03	0.02	[-0.14, 0.15]	54.71%	77.17%	1.004	846.00
Locality_id.L16	-0.18	-0.18	-0.19	[-0.30, -0.07]	99.34%	10.41%	1.017	524.00
Locality_id.L17	-0.13	-0.13	-0.13	[-0.23, -0.04]	98.92%	27.00%	1.005	321.00
Locality_id.L18	-0.15	-0.15	-0.15	[-0.23, -0.07]	99.84%	16.28%	1.007	409.00

Locality_id.L19	-0.04	-0.04	-0.04	[-0.17, 0.10]	67.50%	75.50%	1.006	656.00
Locality_id.L20	-0.16	-0.16	-0.17	[-0.27, -0.07]	99.70%	12.45%	1.005	399.00
Locality_id.L21	0.03	0.03	0.03	[-0.15, 0.23]	61.44%	61.72%	1.001	1668.00
Locality_id.L22	-0.14	-0.14	-0.13	[-0.28, 0.00]	94.61%	31.51%	1.003	635.00
Locality_id.L23	0.07	0.07	0.08	[-0.01, 0.16]	91.34%	70.59%	1.010	437.00
Locality_id.L24	-0.12	-0.12	-0.12	[-0.20, -0.03]	99.35%	35.13%	1.017	443.00
Locality_id.L25	0.04	0.04	5.38e-03	[-0.11, 0.20]	65.20%	70.26%	1.005	934.00
Locality_id.L26	0.20	0.20	0.20	[0.13, 0.27]	100%	0%	1.002	449.00
Locality_id.L27	-0.03	-0.03	-0.02	[-0.12, 0.06]	68.26%	91.91%	1.006	387.00
Locality_id.L28	-0.06	-0.06	-0.06	[-0.12, 0.01]	91.39%	86.26%	1.010	431.00
Locality_id.L29	0.04	0.03	0.04	[-0.11, 0.19]	64.81%	72.32%	1.007	618.00
Locality_id.L30	-0.02	-0.02	-0.01	[-0.19, 0.16]	55.76%	67.91%	1.000	1114.00
Locality_id.L31	-0.13	-0.13	-0.14	[-0.27, 0.01]	93.74%	36.46%	1.008	582.00
Locality_id.L32	0.02	0.02	6.25e-03	[-0.12, 0.17]	56.61%	76.39%	1.004	788.00
Locality_id.L33	0.06	0.06	0.06	[-0.01, 0.14]	89.91%	81.37%	1.008	536.00
Locality_id.L34	0.02	0.02	0.02	[-0.07, 0.10]	61.78%	96.91%	1.004	525.00
Locality_id.L35	-0.23	-0.23	-0.23	[-0.32, -0.15]	100%	0%	1.001	367.00
Locality_id.L36	-0.14	-0.15	-0.14	[-0.24, -0.05]	99.29%	22.36%	1.009	490.00
Locality_id.L37	0.05	0.05	0.05	[-0.07, 0.16]	74.19%	78.13%	1.003	528.00
Locality_id.L38	-0.04	-0.04	-0.03	[-0.17, 0.09]	68.38%	76.75%	1.002	642.00
Locality_id.L39	0.20	0.21	0.19	[0.09, 0.31]	99.94%	3.41%	1.002	568.00
Locality_id.L40	0.06	0.06	0.05	[-0.11, 0.25]	69.35%	59.82%	1.003	1432.00
Locality_id.L41	-0.05	-0.05	-0.07	[-0.19, 0.08]	74.84%	71.84%	1.005	596.00
Locality_id.L42	0.15	0.15	0.15	[0.07, 0.24]	99.85%	13.67%	1.006	391.00
Locality_id.L43	-0.04	-0.04	-0.03	[-0.14, 0.07]	73.29%	83.83%	1.010	500.00
Locality_id.L44	-0.14	-0.15	-0.13	[-0.32, 0.01]	93.50%	31.59%	1.003	748.00
Locality_id.L45	-0.04	-0.04	-0.04	[-0.21, 0.13]	64.84%	65.43%	1.001	1347.00
Locality_id.L46	0.03	0.03	0.03	[-0.11, 0.17]	63.95%	75.54%	1.003	790.00
Locality_id.L47	0.09	0.09	0.09	[-0.06, 0.24]	82.90%	53.20%	1.002	889.00
Locality_id.L48	-0.02	-0.02	-0.03	[-0.14, 0.09]	62.15%	85.83%	1.015	596.00
Locality_id.L49	-0.02	-0.02	-0.02	[-0.16, 0.11]	60.04%	78.59%	1.008	717.00
Locality_id.L50	-4.05e-03	-3.68e-03	-6.39e-03	[-0.08, 0.08]	52.95%	99.18%	1.021	354.00
Locality_id.L51	-0.05	-0.05	-0.05	[-0.22, 0.12]	66.94%	63.39%	1.001	1355.00

Locality_id.L52	-0.10	-0.10	-0.10	[-0.24, 0.06]	86.28%	49.49%	1.002	944.00
Locality_id.L53	-0.06	-0.06	-0.06	[-0.16, 0.03]	85.51%	74.26%	1.003	534.00
Locality_id.L54	6.73e-03	6.58e-03	0.02	[-0.12, 0.13]	53.45%	82.97%	1.007	436.00
Locality_id.L55	0.18	0.18	0.19	[0.02, 0.35]	96.05%	20.51%	1.003	909.00
Locality_id.L56	0.07	0.07	0.06	[-0.01, 0.15]	91.49%	75.05%	1.007	481.00
Locality_id.L57	0.19	0.19	0.19	[0.11, 0.26]	100%	0.55%	1.012	420.00
Locality_id.L58	0.02	0.02	0.02	[-0.06, 0.10]	62.84%	97.29%	1.011	455.00
Locality_id.L59	-0.03	-0.03	-0.03	[-0.21, 0.15]	60.17%	64.68%	1.003	1955.00
Locality_id.L60	-0.06	-0.06	-0.06	[-0.12, 0.01]	92.58%	85.92%	1.004	317.00
Locality_id.L61	-0.12	-0.13	-0.11	[-0.30, 0.04]	89.14%	40.26%	1.004	1145.00
Locality_id.L62	-0.12	-0.12	-0.12	[-0.21, -0.02]	97.39%	37.29%	1.014	408.00
Locality_id.L62	-0.12	-0.12	-0.12	[-0.21, -0.02]	97.39%	37.29%	1.014	408.00
Locality_id.L63	0.10	0.10	0.10	[-0.06, 0.26]	84.67%	49.22%	1.001	911.00
Locality_id.L64	0.11	0.11	0.10	[-0.04, 0.26]	87.81%	45.58%	1.004	678.00
Locality_id.L65	0.12	0.12	0.11	[-0.01, 0.23]	93.90%	41.49%	1.007	660.00
Locality_id.L66	0.04	0.04	0.04	[-0.08, 0.15]	69.35%	82.36%	1.001	505.00
Locality_id.L67	0.07	0.07	0.06	[0.01, 0.14]	95.71%	75.32%	1.019	375.00
Locality_id.L68	0.01	0.02	0.01	[-0.05, 0.08]	65.12%	100%	1.007	393.00
(Intercept)	0.13	0.13	0.13	[0.10, 0.16]	100%	0%	1.008	534.00

Random Effects (Count Model): nace_section_letter

Parameter	Median	Mean	MAP	89% CI	pd	% in ROPE	Rhat	ESS
nace_section_letter.A	-0.31	-0.31	-0.31	[-0.44, -0.17]	99.98%	0%	1.003	643.00
nace_section_letter.B	-5.18e-04	-3.36e-03	0.03	[-0.26, 0.26]	50.20%	47.12%	1.005	1024.00
nace_section_letter.C	0.18	0.18	0.19	[0.06, 0.31]	98.86%	11.47%	1.004	644.00
nace_section_letter.D	0.06	0.05	0.06	[-0.13, 0.25]	67.61%	58.30%	1.002	599.00
nace_section_letter.E	0.37	0.37	0.35	[0.19, 0.56]	99.95%	0%	1.003	649.00
nace_section_letter.F	-0.57	-0.56	-0.57	[-0.69, -0.43]	100%	0%	1.007	684.00
nace_section_letter.G	-0.12	-0.12	-0.11	[-0.24, 0.00]	94.88%	37.28%	1.006	729.00
nace_section_letter.H	-0.30	-0.30	-0.31	[-0.43, -0.16]	99.95%	0%	1.005	643.00

nace_section_letter.I	0.52	0.52	0.53	[0.40, 0.65]	100%	0%	1.003	690.00
nace_section_letter.J	0.40	0.40	0.42	[0.26, 0.53]	100%	0%	1.011	603.00
nace_section_letter.K	0.23	0.23	0.25	[0.09, 0.38]	99.45%	4.55%	1.005	690.00
nace_section_letter.L	0.09	0.09	0.09	[-0.04, 0.22]	86.49%	54.92%	1.004	662.00
nace_section_letter.M	-0.26	-0.26	-0.26	[-0.39, -0.13]	99.92%	0%	1.005	714.00
nace_section_letter.N	-0.28	-0.28	-0.27	[-0.41, -0.14]	99.96%	0%	1.007	587.00
nace_section_letter.O	0.08	0.08	0.07	[-0.04, 0.22]	84.24%	58.96%	1.002	666.00
nace_section_letter.P	0.02	0.02	1.43e-03	[-0.13, 0.18]	56.57%	73.26%	1.006	640.00
nace_section_letter.Q	-0.04	-0.04	-0.04	[-0.20, 0.11]	67.97%	69.00%	1.013	559.00
nace_section_letter.R	0.17	0.17	0.18	[0.02, 0.31]	96.49%	21.42%	1.010	590.00
nace_section_letter.S	-0.35	-0.36	-0.35	[-0.49, -0.22]	100%	0%	1.007	721.00
nace_section_letter.T	0.07	0.07	0.06	[-0.37, 0.52]	60.61%	28.20%	1.002	4210.00
(Intercept)	0.30	0.31	0.28	[0.22, 0.39]	100%	0%	1.002	1481.00

Model 2B

This section presents the R outputs corresponding to Model 2B. The following presents the summary of Model 2B.

```
Family: gaussian
  Links: mu = identity
Formula: log_consumption_std ~ Year + Company_Size_z + (1 | Company_id) + (1 | nace_section_letter) + (1 | Locality_id)
   Data: db (Number of observations: 55374)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 8000

Multilevel Hyperparameters:
~Company_id (Number of levels: 11706)
                 Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)       0.88       0.01      0.87      0.89  1.00       491      1280

~Locality_id (Number of levels: 68)
                 Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)       0.13       0.02      0.10      0.17  1.01       408       766

~nace_section_letter (Number of levels: 20)
                 Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)       0.31       0.05      0.23      0.43  1.00      1163      1635

Regression Coefficients:
                 Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
Intercept           18.83       2.11     14.67     22.95  1.00     10332      5235
Year                -0.01       0.00     -0.01     -0.01  1.00     10503      5134
Company_Size_z       0.18       0.01      0.17      0.20  1.00      1466      2786

Further Distributional Parameters:
      Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sigma     0.33       0.00      0.33      0.34  1.00      8793      6258
```

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

The following summarise key descriptive statistics of the posterior distribution for the parameters of Model 2B. To maintain clarity and interpretability, for the company-level random intercepts, we present the five smallest companies and the five largest companies, based on the company size variable.

Fixed Effects

Parameter	Median	Mean	MAP	89% CI	pd	% in ROPE	Rhat	ESS
(Intercept)	18.82	18.83	18.68	[15.51, 22.18]	100%	0%	1.000	10262.00
Year	-9.31e-03	-9.31e-03	-9.30e-03	[-0.01, -0.01]	100%	100%	1.000	10439.00
Company_Size_z	0.18	0.18	0.18	[0.17, 0.19]	100%	0%	1.001	1456.00

Sigma

Parameter	Median	Mean	MAP	89% CI	pd	% in ROPE	Rhat	ESS
sigma	0.33	0.33	0.33	[0.33, 0.34]	100%	0%	1.000	8722.00

Random Effects (Count Model): Company_id

Parameter	Median	Mean	MAP	89% CI	pd	% in ROPE	Rhat	ESS
Company_id.5	0.52	0.52	0.52	[0.23, 0.78]	99.94%	0%	1.002	1633.00
Company_id.7	-0.94	-0.94	-0.95	[-1.22, -0.68]	100%	0%	1.002	1246.00
Company_id.48	-0.22	-0.22	-0.23	[-0.48, 0.05]	91.06%	21.24%	1.005	1309.00

Company_id.49	1.54	1.55	1.52	[1.26, 1.81]	100%	0%	1.002	2401.00
Company_id.62	-0.44	-0.44	-0.43	[-0.69, -0.16]	99.56%	0%	1.007	1199.00
Company_id.49838	-0.69	-0.69	-0.66	[-0.98, -0.41]	100%	0%	1.003	1766.00
Company_id.24514	1.03	1.02	1.05	[0.69, 1.37]	100%	0%	1.001	1763.00
Company_id.44233	-3.04	-3.04	-3.04	[-3.41, -2.70]	100%	0%	1.002	1275.00
Company_id.12308	-1.37	-1.37	-1.35	[-1.75, -1.00]	100%	0%	1.002	1439.00
Company_id.58635	-14.46	-14.45	-14.50	[-15.49, -13.42]	100%	0%	1.001	1481.00
(Intercept)	0.88	0.88	0.88	[0.87, 0.89]	100%	0%	1.004	475.00

Random Effects (Count Model): Locality_id

Parameter	Median	Mean	MAP	89% CI	pd	% in ROPE	Rhat	ESS
Locality_id.L01	0.01	0.02	8.30e-03	[-0.05, 0.08]	64.89%	100%	1.004	317.00
Locality_id.L02	0.26	0.26	0.26	[0.17, 0.36]	100%	0%	1.023	364.00
Locality_id.L03	-0.04	-0.04	-0.02	[-0.18, 0.09]	69.99%	73.68%	1.017	502.00
Locality_id.L04	-0.07	-0.07	-0.08	[-0.18, 0.03]	84.89%	68.24%	1.028	253.00
Locality_id.L05	-7.97e-03	-6.96e-03	-0.02	[-0.14, 0.12]	53.29%	81.09%	1.006	531.00
Locality_id.L06	0.13	0.13	0.13	[0.03, 0.22]	98.47%	30.70%	1.014	354.00
Locality_id.L07	-0.02	-0.01	-0.02	[-0.16, 0.12]	56.89%	76.71%	1.016	357.00
Locality_id.L08	-0.04	-0.04	-0.03	[-0.18, 0.10]	69.69%	72.72%	1.005	580.00
Locality_id.L09	-0.02	-0.02	-0.02	[-0.16, 0.12]	58.79%	77.50%	1.010	493.00
Locality_id.L10	-0.04	-0.04	-0.03	[-0.16, 0.07]	70.19%	79.62%	1.008	336.00
Locality_id.L11	0.10	0.10	0.07	[-0.07, 0.26]	83.49%	51.22%	1.007	627.00
Locality_id.L12	0.14	0.14	0.13	[-0.01, 0.30]	93.12%	33.64%	1.003	502.00
Locality_id.L13	0.15	0.15	0.16	[0.06, 0.24]	99.16%	17.54%	1.009	292.00
Locality_id.L14	-0.12	-0.12	-0.12	[-0.18, -0.06]	99.99%	26.72%	1.007	340.00
Locality_id.L15	6.96e-03	6.12e-03	2.06e-03	[-0.14, 0.15]	53.15%	76.58%	1.005	652.00
Locality_id.L16	-0.18	-0.19	-0.19	[-0.30, -0.07]	99.52%	10.36%	1.006	396.00
Locality_id.L17	-0.12	-0.13	-0.12	[-0.22, -0.03]	98.75%	33.11%	1.005	377.00
Locality_id.L18	-0.15	-0.15	-0.15	[-0.22, -0.06]	99.90%	15.25%	1.008	329.00

Locality_id.L19	-0.04	-0.04	-0.03	[-0.18, 0.11]	67.84%	73.26%	1.015	458.00
Locality_id.L20	-0.18	-0.18	-0.17	[-0.27, -0.08]	99.56%	8.84%	1.009	322.00
Locality_id.L21	0.03	0.04	0.02	[-0.15, 0.23]	61.58%	61.03%	1.006	1325.00
Locality_id.L22	-0.14	-0.14	-0.13	[-0.28, 0.00]	95.36%	32.45%	1.006	552.00
Locality_id.L23	0.07	0.07	0.06	[-0.02, 0.15]	90.85%	70.22%	1.005	359.00
Locality_id.L24	-0.12	-0.12	-0.12	[-0.20, -0.05]	99.44%	33.11%	1.021	320.00
Locality_id.L25	0.05	0.04	0.06	[-0.11, 0.20]	67.34%	68.58%	1.003	651.00
Locality_id.L26	0.20	0.21	0.20	[0.13, 0.28]	100%	0%	1.009	360.00
Locality_id.L27	-0.03	-0.03	-0.04	[-0.12, 0.06]	71.03%	92.36%	1.005	331.00
Locality_id.L28	-0.06	-0.06	-0.06	[-0.13, 0.01]	92.06%	85.34%	1.006	345.00
Locality_id.L28	-0.06	-0.06	-0.06	[-0.13, 0.01]	92.06%	85.34%	1.006	345.00
Locality_id.L29	0.04	0.04	0.05	[-0.10, 0.19]	67.14%	71.39%	1.008	616.00
Locality_id.L30	-0.02	-0.02	-0.02	[-0.19, 0.15]	55.80%	68.66%	1.004	918.00
Locality_id.L31	-0.13	-0.13	-0.13	[-0.27, 0.00]	94.09%	35.47%	1.011	579.00
Locality_id.L32	0.02	0.02	0.02	[-0.13, 0.17]	57.45%	74.68%	1.008	473.00
Locality_id.L33	0.06	0.06	0.07	[-0.02, 0.14]	89.29%	79.47%	1.001	365.00
Locality_id.L34	0.02	0.02	0.03	[-0.06, 0.10]	62.88%	96.72%	1.013	319.00
Locality_id.L35	-0.23	-0.23	-0.24	[-0.31, -0.14]	100%	0%	1.002	377.00
Locality_id.L36	-0.15	-0.15	-0.14	[-0.25, -0.05]	99.17%	20.24%	1.005	367.00
Locality_id.L37	0.04	0.04	0.05	[-0.06, 0.15]	73.22%	80.53%	1.014	374.00
Locality_id.L38	-0.03	-0.03	-0.02	[-0.16, 0.08]	66.06%	79.71%	1.006	445.00
Locality_id.L39	0.20	0.21	0.20	[0.09, 0.31]	99.89%	3.49%	1.004	393.00
Locality_id.L40	0.06	0.06	0.06	[-0.11, 0.24]	70.03%	61.43%	1.011	782.00
Locality_id.L41	-0.06	-0.06	-0.06	[-0.18, 0.07]	77.58%	71.25%	1.004	491.00
Locality_id.L42	0.16	0.16	0.17	[0.07, 0.25]	99.72%	13.86%	1.008	295.00
Locality_id.L43	-0.04	-0.04	-0.05	[-0.15, 0.06]	74.38%	82.29%	1.008	451.00
Locality_id.L44	-0.15	-0.15	-0.15	[-0.31, 0.01]	93.76%	29.21%	1.003	734.00
Locality_id.L45	-0.04	-0.04	-0.04	[-0.22, 0.12]	63.39%	65.82%	1.006	956.00
Locality_id.L46	0.04	0.04	0.04	[-0.11, 0.18]	65.41%	73.84%	1.013	493.00
Locality_id.L47	0.09	0.09	0.09	[-0.07, 0.23]	83.33%	55.09%	1.003	667.00
Locality_id.L48	-0.02	-0.02	-3.16e-03	[-0.15, 0.09]	60.36%	83.97%	1.014	408.00
Locality_id.L49	-0.02	-0.02	-0.04	[-0.17, 0.11]	60.38%	77.03%	1.007	423.00
Locality_id.L50	-5.40e-03	-6.60e-03	-6.00e-03	[-0.09, 0.08]	54.43%	98.46%	1.004	335.00

Locality_id.L51	-0.05	-0.05	-0.05	[-0.22, 0.12]	67.71%	63.54%	1.001	1010.00
Locality_id.L52	-0.10	-0.10	-0.10	[-0.26, 0.06]	85.32%	49.61%	1.007	729.00
Locality_id.L53	-0.06	-0.06	-0.06	[-0.16, 0.04]	83.29%	74.32%	1.013	401.00
Locality_id.L54	-1.97e-03	9.29e-04	-0.02	[-0.13, 0.12]	51.12%	83.20%	1.008	489.00
Locality_id.L55	0.18	0.18	0.17	[0.01, 0.34]	96.61%	20.39%	1.002	571.00
Locality_id.L56	0.07	0.07	0.07	[-0.01, 0.15]	90.91%	73.62%	1.010	342.00
Locality_id.L57	0.19	0.19	0.19	[0.11, 0.26]	100%	0.12%	1.012	346.00
Locality_id.L58	0.02	0.02	0.02	[-0.06, 0.09]	64.48%	98.29%	1.014	242.00
Locality_id.L59	-0.03	-0.03	-0.03	[-0.21, 0.16]	59.39%	63.76%	1.005	1262.00
Locality_id.L60	-0.06	-0.06	-0.06	[-0.12, 0.00]	92.90%	87.80%	1.013	291.00
Locality_id.L61	-0.13	-0.13	-0.13	[-0.29, 0.03]	90.09%	38.74%	1.006	623.00
Locality_id.L62	-0.12	-0.12	-0.12	[-0.21, -0.02]	97.90%	38.99%	1.005	358.00
Locality_id.L63	0.10	0.11	0.09	[-0.07, 0.27]	84.49%	48.95%	1.007	664.00
Locality_id.L64	0.11	0.11	0.10	[-0.03, 0.27]	88.12%	47.82%	1.006	752.00
Locality_id.L65	0.12	0.12	0.12	[-0.01, 0.25]	92.79%	41.64%	1.013	365.00
Locality_id.L66	0.04	0.04	0.05	[-0.08, 0.15]	70.84%	81.01%	1.006	501.00
Locality_id.L67	0.08	0.07	0.08	[0.01, 0.15]	95.51%	72.12%	1.011	301.00
Locality_id.L68	0.01	0.01	0.02	[-0.05, 0.08]	63.61%	100%	1.014	283.00
(Intercept)	0.13	0.13	0.13	[0.11, 0.16]	100%	0%	1.008	417.00

Random Effects (Count Model): nace_section_letter

Parameter	Median	Mean	MAP	89% CI	pd	% in ROPE	Rhat	ESS
nace_section_letter.A	-0.31	-0.31	-0.29	[-0.44, -0.19]	100%	0%	1.005	561.00
nace_section_letter.B	-4.38e-03	-3.84e-03	-0.02	[-0.27, 0.27]	51.12%	47.34%	1.010	715.00
nace_section_letter.C	0.19	0.19	0.18	[0.06, 0.30]	99.35%	9.55%	1.008	574.00
nace_section_letter.D	0.06	0.06	0.06	[-0.12, 0.25]	70.58%	58.01%	1.011	454.00
nace_section_letter.E	0.38	0.38	0.37	[0.21, 0.57]	99.96%	0%	1.007	515.00
nace_section_letter.F	-0.56	-0.56	-0.57	[-0.69, -0.43]	100%	0%	1.012	508.00
nace_section_letter.G	-0.12	-0.12	-0.11	[-0.24, 0.00]	94.75%	39.78%	1.005	629.00
nace_section_letter.H	-0.30	-0.30	-0.29	[-0.43, -0.16]	99.98%	0%	1.020	402.00

nace_section_letter.I	0.53	0.53	0.53	[0.40, 0.65]	100%	0%	1.005	585.00
nace_section_letter.J	0.41	0.40	0.41	[0.28, 0.53]	99.98%	0%	1.007	519.00
nace_section_letter.K	0.24	0.24	0.22	[0.10, 0.38]	99.72%	3.50%	1.005	419.00
nace_section_letter.L	0.09	0.09	0.11	[-0.04, 0.22]	86.65%	53.45%	1.001	481.00
nace_section_letter.M	-0.26	-0.26	-0.27	[-0.40, -0.13]	99.88%	0.43%	1.011	495.00
nace_section_letter.N	-0.28	-0.28	-0.27	[-0.41, -0.14]	99.99%	0%	1.007	529.00
nace_section_letter.O	0.09	0.09	0.09	[-0.06, 0.20]	85.30%	56.87%	1.004	570.00
nace_section_letter.P	0.02	0.02	0.02	[-0.13, 0.17]	57.56%	72.91%	1.008	520.00
nace_section_letter.Q	-0.04	-0.04	-0.04	[-0.20, 0.11]	65.35%	71.17%	1.011	421.00
nace_section_letter.R	0.17	0.17	0.19	[0.02, 0.31]	96.80%	20.41%	1.002	435.00
nace_section_letter.S	-0.34	-0.35	-0.34	[-0.48, -0.22]	100%	0%	1.009	585.00
nace_section_letter.T	0.06	0.07	0.05	[-0.37, 0.51]	59.31%	29.07%	1.000	3262.00
(Intercept)	0.30	0.31	0.29	[0.23, 0.39]	100%	0%	1.003	1269.00

Model 3

This section presents the R outputs corresponding to Model 3. The following presents the summary of Model 3.

```
Family: gaussian
  Links: mu = identity
Formula: log_consumption_std ~ Year + Company_Size_z + (1 | Company_id) + (1 | nace_section_letter) + (1 | Locality_id) + (1 |
nace_section_letter:Year) + (1 | nace_section_letter:Locality_id) + (1 | Year:Locality_id)
   Data: db (Number of observations: 55374)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 8000

Multilevel Hyperparameters:
~Company_id (Number of levels: 11706)
                 Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)        0.85       0.01      0.84      0.86  1.00       764      1679

~Locality_id (Number of levels: 68)
                 Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)        0.10       0.02      0.05      0.14  1.01       617       747

~nace_section_letter (Number of levels: 20)
                 Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)        0.31       0.06      0.22      0.44  1.00      1588      2677

~nace_section_letter:Locality_id (Number of levels: 982)
                 Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)        0.25       0.02      0.22      0.28  1.00       425       773

~nace_section_letter:Year (Number of levels: 100)
                 Estimate  Est.Error  l-95% CI  u-95% CI  Rhat  Bulk_ESS  Tail_ESS
sd(Intercept)        0.04       0.00      0.03      0.05  1.00      2039      4269
```

~Year:Locality_id (Number of levels: 340)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.03	0.00	0.03	0.04	1.00	3812	5565

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	17.16	7.32	3.18	31.33	1.00	2756	3975
Year	-0.01	0.00	-0.02	-0.00	1.00	2753	4021
Company_Size_z	0.18	0.01	0.16	0.19	1.00	2132	3515

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.33	0.00	0.33	0.33	1.00	8838	6000

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

The following summarise key descriptive statistics of the posterior distribution for the parameters of the interaction terms included in Model 3.

Summary of Posterior Distribution SD/Cor: Company_id

Parameter	Median	89% CI	pd	Rhat	ESS
(Intercept)	0.85	[0.84, 0.86]	100%	1.001	756.00

Random effects SD/Cor: Locality_id

Parameter	Median	89% CI	pd	Rhat	ESS
(Intercept)	0.10	[0.06, 0.13]	100%	1.010	577.00

Random effects SD/Cor: nace_section_letter

Parameter	Median	89% CI	pd	Rhat	ESS
(Intercept)	0.30	[0.22, 0.39]	100%	1.000	1655.00

Random effects SD/Cor: nace_section_letter:Locality_id

Parameter	Median	89% CI	pd	Rhat	ESS
(Intercept)	0.25	[0.22, 0.27]	100%	1.005	418.00

Random effects SD/Cor: nace_section_letter:Year

Parameter	Median	89% CI	pd	Rhat	ESS
(Intercept)	0.04	[0.03, 0.05]	100%	1.001	2036.00

Random effects SD/Cor: Year:Locality_id

Parameter	Median	89% CI	pd	Rhat	ESS
(Intercept)	0.03	[0.03, 0.03]	100%	1.000	3786.00

Posterior summaries for all model parameters were obtained. The output was subsequently filtered to retain parameters corresponding to standard deviations (identified by the 'sd_' prefix), in order to evaluate the variability associated with the interaction (random) effects.

Summary of Posterior Distribution SD/Cor: Company_id

Parameter	Median	89% CI	pd	Rhat	ESS
(Intercept)	0.85	[0.84, 0.86]	100%	1.001	756.00

Random effects SD/Cor: Locality_id

Parameter	Median	89% CI	pd	Rhat	ESS
(Intercept)	0.10	[0.06, 0.13]	100%	1.010	577.00

Random effects SD/Cor: nace_section_letter

Parameter	Median	89% CI	pd	Rhat	ESS
(Intercept)	0.30	[0.22, 0.39]	100%	1.000	1655.00

Random effects SD/Cor: nace_section_letter:Locality_id

Parameter	Median	89% CI	pd	Rhat	ESS
(Intercept)	0.25	[0.22, 0.27]	100%	1.005	418.00

Random effects SD/Cor: nace_section_letter:Year

Parameter	Median	89% CI	pd	Rhat	ESS
(Intercept)	0.04	[0.03, 0.05]	100%	1.001	2036.00

Random effects SD/Cor: Year:Locality_id

Parameter	Median	89% CI	pd	Rhat	ESS
(Intercept)	0.03	[0.03, 0.03]	100%	1.000	3786.00